

Aryamaan Dash

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EDUCATION

University of California, Irvine

Bachelor of Science in Computer Science & Engineering, Minor in Mathematics

Clubs: AI at UCI, IEEE at UCI, ACM at UCI, Engineering Student Council

Irvine, CA

June 2028

EXPERIENCE

Software Engineer Intern

March 2024 – December 2024

Hey, Blue!

Remote

- Built and shipped 6 responsive pages and reusable UI components using **React**, **JavaScript**, **HTML**, and **CSS**, supporting consistent desktop and mobile experiences.
- Integrated **Supabase** for database-backed application features and client-side data access, connecting the React front end to persistent backend services.
- Led a 5-developer front-end team using **Git/GitHub**, translating product requirements into scoped tickets, coordinating feature ownership, and driving changes through code review and release.

Learning Assistant

Sep. 2026 – Present

UC Irvine - ICS

Irvine, CA

- Support students in an introductory **Python** course focused on software libraries, helping debug programs, interpret errors, and apply library APIs correctly in programming assignments.
- Guide students through **software development** including modular design, data structures, testing, and decomposition of larger programming problems into maintainable components.
- Provide technical feedback during labs and office hours, reviewing student code for **correctness**, **readability**, and **debugging strategy** while reinforcing effective problem-solving practices.

PROJECTS

Study Tracker | *Next.js, TypeScript, Auth.js, Prisma, PostgreSQL, Docker*

Apr. 2026 – Sep. 2026

- Built and deployed a **full-stack web application** using **Next.js**, **TypeScript**, **PostgreSQL**, **Prisma**, and **Auth.js**, implementing authenticated user sessions and persistent study data.
- Developed end-to-end product features including interactive charts, calendars, focus timers, journal entries, and weekly reflections, integrating **frontend components with backend data workflows**.
- Improved application performance through **parallel data fetching**, **indexed database queries**, and **tag-based caching**; added automated testing and **Dockerized development** workflows.

CPU Optimized Inference Engine | *C++20, Cmake, ARM NEON, Apple Accelerate*

Aug. 2026 – Sep. 2026

- Built a **C++20 CPU inference engine** supporting dense layers, convolution, and multilayer perceptrons with **FP32 and INT8 quantized execution**.
- Accelerated matrix multiplication through **cache tiling**, **ARM NEON SIMD**, and **multithreading**, optimizing inference for batch sizes of 1 and larger workloads.
- Implemented **INT8 kernels with 32-bit accumulation** and reproducible benchmarks against **Apple Accelerate**, evaluating numerical accuracy, throughput, and end-to-end latency.

Programmable Multi-Effects Guitar Pedal | *C++, STM32, DaisySP, KiCad*

Jun. 2026 – Sep. 2026

- Designed and built an **embedded hardware-software system** around an STM32-based Daisy Seed, integrating custom circuitry and firmware to support five selectable real-time guitar effects.
- Developed **real-time C++ firmware** using audio callbacks, DSP processing, delay-line buffers, wet/dry mixing, and debounced input handling while maintaining continuous low-latency audio processing.
- Designed and validated the pedal's circuit and **custom PCB in KiCad**, debugging hardware-software integration issues and testing the completed system with **15 users**.

TECHNICAL SKILLS

Languages: C/C++, Python, SQL (Postgres), TypeScript, HTML/CSS, Verilog

Frameworks: PyTorch, Tensorflow, Flask, FastAPI, Next.js, Node.js

Developer Tools: Git, Github Actions, Docker, CMake, AWS, Google ADK, VS Code, Linux, Vivado, STM32CubeMX

Libraries: scikit-learn, NumPy, pandas, OpenCV, Librosa, Matplotlib, Keras, React